Simplify algebraic expressions

1 Simplify the following:

a $(5u - 2u) \times 3$

b $(13a + 15a) \div 7$

c $9p \div (6p - 3p)$

d $80mn \div (4m) \div (5n).$

e $(-7n) \times 4 - (2n + 5n)$

f $(6u \times 12vw) \div (6v \times 32u)$

g $(19v - 5v) \div 2 \times 3.$

h $30r \div (3r + 2r) + 5r.$

i $5v - 5\,(6v - 2v)$

j $6v + (15v - (9v - 2v)).$

k $-8t - (8t - 6t \div 2)$

l $((-8r) \times 16st) \div (24s \times (-9r))$

m $(7m - 5m) \times 4m$

n $(-4n) \times 8 - (7n + 4n)$

2 Simplify the following:

a $\dfrac{16x}{3x + 5x}$

b $\dfrac{20s}{4s} \times 6t$

c $\dfrac{21m}{3} + \dfrac{10m}{2}$

d $9v \times 2 - \dfrac{25v}{5}$

e $15m + \dfrac{51m - m}{5m}.$

f $\dfrac{58r + 5r}{7r \times 3}$

g $\dfrac{7r + 5r}{7r - 5r}$

h $\dfrac{7rs + 23rs}{5r \times 3s}$

i $5n^2 + \dfrac{20n^2}{5}$

j $3m - \dfrac{12m^2}{3m}.$

k $5n \times 4n - \dfrac{8n^2}{4}$

l $\dfrac{7st + 53st}{4s \times 5t}$

3 Simplify the following:

a $2t + 4 \times 8t$

b $2c + 7c \times 3 + 4c$

c $61a - 6 \times 9a - 2a$

d $8 \times 3a + 6 \times 2a$

e $3n^2 + 8n^2 - 3n^2$

f $9n^2 + 3n \times 6n$

g $5n^2 + 5n^2 \times 2 + 7n^2$

h $5mn \times 3n - 2mn^2$

i $20w - 10w \div 2 - 3w$

j $6p \times 8q \div 2$

k $60n^2 \div 5n \div 3$

l $8jk \times 15k \div 10j$

m $20pq \div 5q \times 2p$

n $12s - 8s \div 4$

o $12u \div 3 - 2u$

p $8jk \times 15k \div 20j$

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Algebraic fractions

4 Consider $\frac{w}{3} + \frac{w}{9}$.

a Find the lowest common denominator of $\frac{w}{3}$ and $\frac{w}{9}$.

b Hence, write $\frac{w}{3} + \frac{w}{9}$ as a single fraction.

5 Consider $\frac{m}{4} - \frac{m}{12}$.

a Find the lowest common denominator of $\frac{m}{4}$ and $\frac{m}{12}$.

b Hence, write $\frac{m}{4} - \frac{m}{12}$ as a single fraction in simplest form.

6 Simplify the following expressions:

a $\frac{8x}{12} + \frac{10x}{12}$ **b** $\frac{y}{11} - \frac{9y}{11}$ **c** $\frac{17y}{20} + \frac{14y}{20}$ **d** $\frac{7x}{3} - \frac{5x}{3}$

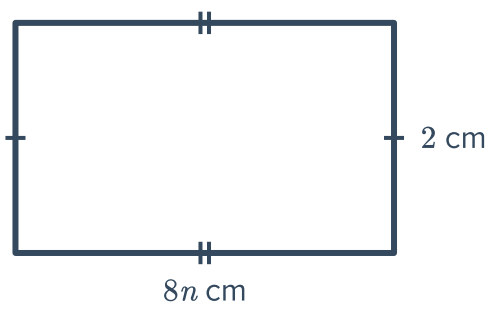
e $\frac{3x}{4} + \frac{3x+4}{4}$ **f** $\frac{p}{2} - \frac{p}{4}$ **g** $\frac{3p}{7} + \frac{2p}{35}$ **h** $\frac{p}{3} + \frac{6p}{15}$

i $\frac{x}{3} + \frac{x+2}{12}$ **j** $\frac{4x+3}{3} - \frac{x-3}{6}$

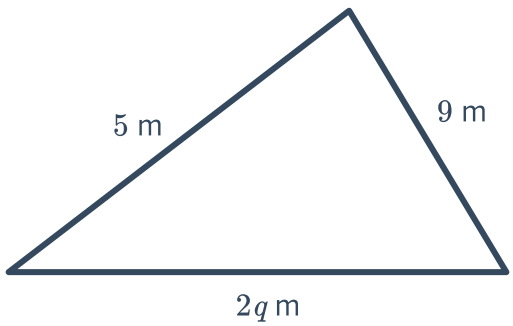
Applications

7 Write an algebraic expression for the perimeter of the following shapes:

a

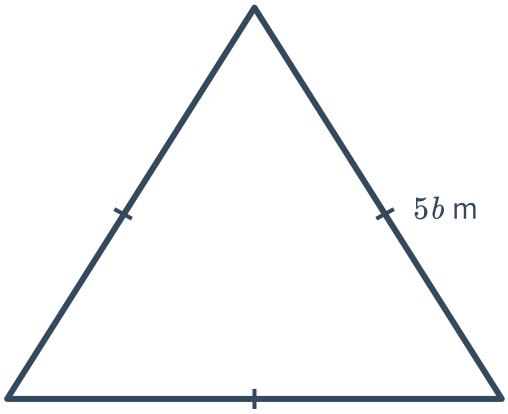


b

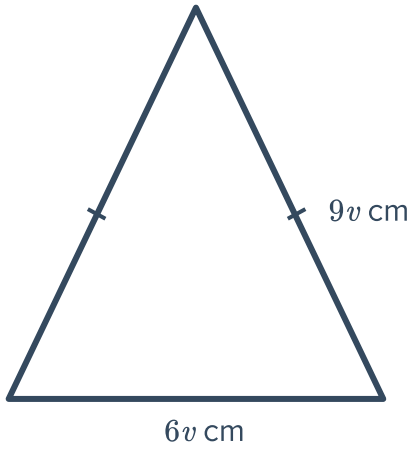


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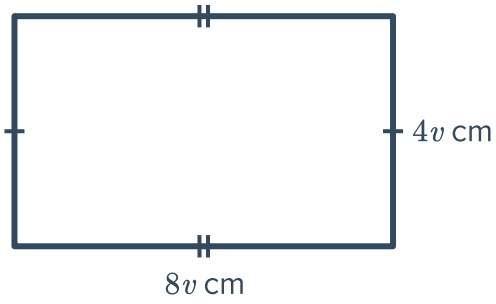
c



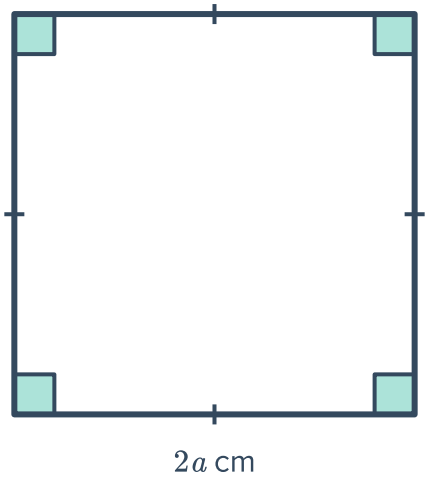
d



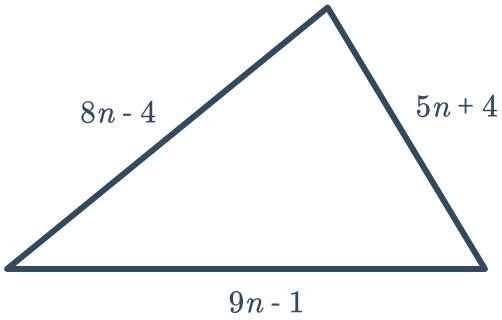
e



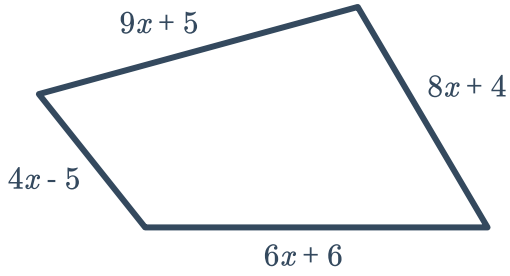
f



g

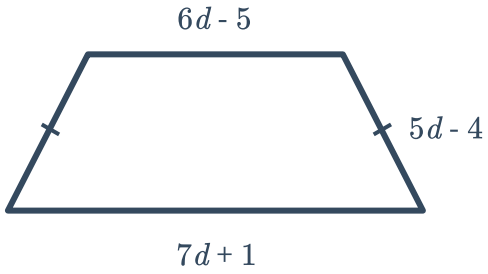


h



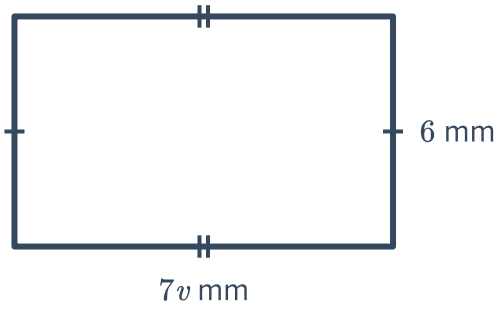
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i

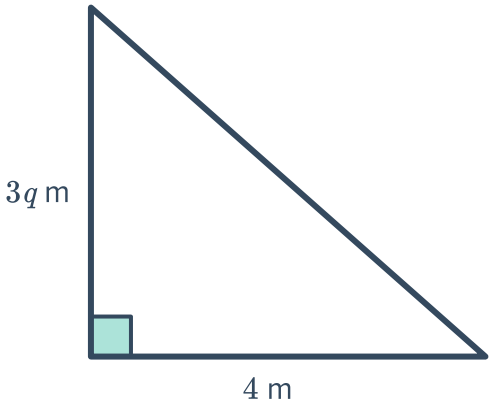


8 Write an algebraic expression for the area of the following shapes:

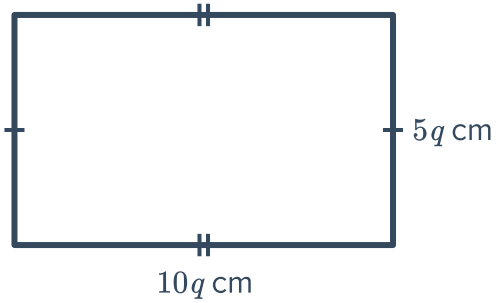
a



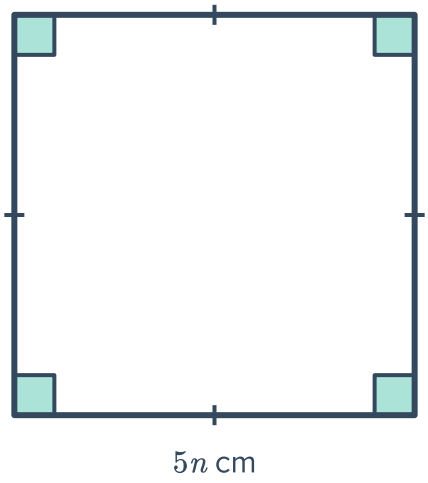
b



c



d

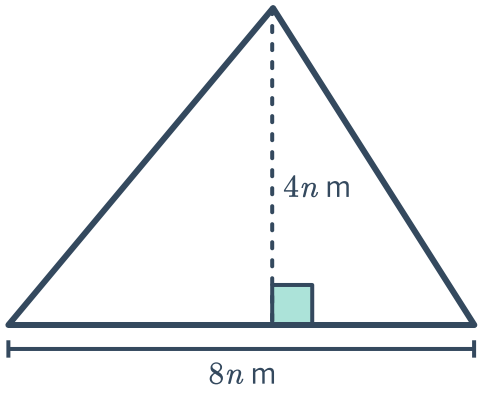


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e



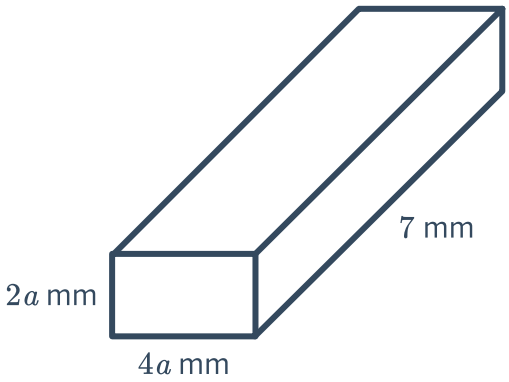
f



9 Consider the following rectangular prism:

a Write an expression for its volume.

b Write an expression for its surface area.



10 Dylan, Jimmy and Valentina are travelling from Adelaide to Sydney at x, y and z km/h respectively.

Let S_1 represent the average speed of Dylan and Jimmy's vehicles, and S_2 represent the average speed of Dylan and Valentina's vehicles.

Write a simplified expression for $S_1 + S_2$ in terms of x, y and z.

11 Frank wants to determine the profit he makes on each eraser he sells. Frank can sell 3 erasers for x dollars, and it costs y dollars to make 9 erasers.

a Determine a simplified expression for the profit Frank makes on each eraser. Leave your answer as a single fraction.

b Frank realised he could sell 3 erasers for $\$3$ more than they previously were, and reduce the cost of production of 9 erasers by $\$2$. Find the new profit per eraser.

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12 Glen lived in a rectangular bedroom. His mother asked him to find out the area of his room, and since Glen did not have a measuring tape or ruler, he counted how many of his steps each wall was.

If Glen's foot length is x cm, and two adjacent walls were 12 and 10 footsteps long, write an expression for the area of his room.

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